

Tvarita Mathur

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EDUCATION

Fairview High School

Boulder, CO

W GPA: 4.69/5.0 UW GPA: 3.773/4.0

Expected Graduation : 2027

Relevant Coursework already completed: Pre IB Biology, IB Chemistry, AP Computer Science Principles (Python), AP Computer Science A(Java), AP Calculus AB, AP Calculus BC, IB Physics, AP Statistics, AP Language and Composition, AP Psychology, IB Spanish 4

Current year courses: AP Biology, IB Spanish 5 HL, AP Physics C, AP Literature and Composition, Calculus 2/3

RESEARCH & LABORATORY EXPERIENCE

Dr. Vernery's Soft Matter Mechanics Lab, CU Boulder

May 2025- present

- Conducted independent research on fungal hyphae growth behaviour, developing original hypotheses about environmental factors affecting growth patterns
- Utilized laboratory techniques including sterile cultivation methods, micropipetting, and precision microscopy for quantitative growth measurements
- Captured and analyzed microscopy images to quantify growth, demonstrating proficiency in scientific documentation and data collection

Dr. DeSouza's Integrative Vascular Biology Laboratory, CU Boulder

May 2025-present

- Performed advanced cellular manipulation techniques in controlled environments, including feeding, splitting and pharmacological treatment of cell cultures under fume hood conditions
- Conducted statistical analysis on experimental data to identify significant patterns and physiological responses

Independent Science Fair Research Project

October 2025-present

- Developed original hypothesis studying applications of transient network theory to biological systems
- Designed and executed independent experiments in laboratory setting, demonstrating ability to translate theoretical concepts into testable hypotheses
- Performed statistical analysis on experimental data and drew evidence-based conclusions presenting findings in formal science fair format

STEM LEADERSHIP & COMPETITION

Fairview Formula 1 club

2024 - Present

Founder & President

- Founded integrating physics, engineering, and aerodynamic principles with real-world racing applications
- Organize STEM learning activities focused on materials science, aerodynamics and biomechanics in motorsport
- Led fundraising efforts for Mission 44, promoting equal STEM opportunities for students worldwide

Fairview Science Olympiad

2022- Present

Co-President (present) Vice President(2025-2026), Captain (2024-2025)

- 4th place in Anatomy and Physiology at state level** - demonstrated mastery of human body systems, physiological processes, and medical terminology
- Led team to 6th place overall team finish at 2025 and 2026 state competition
- Led team preparation sessions on cellular biology, genetics, and anatomical systems

Johns Hopkins Global Health Leadership Conference

July 2026

- Explored global health challenges through learning from public health researchers at Johns Hopkins University
- Presented independent research on improving access to medical adhesives

Altona Middle School Science Olympiad

2022-Present

Volunteer

- Coached 60 middle school Science Olympiad students to 4th at state level, taught event content, and helped the team prepare for competition.

Mathnasium Learning Center

2021-2024

Volunteer

- Tutored students in mathematics, statistics, and quantitative analysis skills essential for scientific research
- Adapted teaching methods in diverse learning styles, developing communication skills valuable for collaborative research

SPORTS

Fairview Track & Field

2021-Present

- Balanced competitive athletics with academic excellence, exemplifying discipline and time management

ADVOCACY & LEADERSHIP

Representative Manny Rutinel, Campaign for Colorado's 8th Congressional District
Intern

June 2026-Present

- Made 3,000+ voter outreach calls and canvassed 200+ doors while supporting campaign communications and social media for a U.S. Congressional campaign.

Representative Eliza Hamrick, Colorado General Assembly
Research Intern

Feb 2025-Present

- Conducting research and draft legislative improvements to the Colorado READ Act for better accommodations for testing students with reading disabilities.
- Researched Colorado policies for teacher evaluations and came up with changes to legislation on teacher evaluations across the state.
- Researched and proposed legislation for state funded flexible day care systems catered towards front line workers.

Colorado Gifted and Talented Student Board

2024-Present

- Drafted and proposed education legislation to state legislators at the Colorado State Capitol, advocating for an improved GT program through removing teacher recommendations for GT testing and increasing funding for resources towards the GT program especially in Title I schools.
- Drafted testimonies for education reform bills to support their passage, developing public speaking and policy advocacy skills

BVSD Gifted and Talented Youth Council

Leader

2023-present

- Collaborated with district administration to improve gifted program
- Worked with school administration to initiate and improve the GT program at Fairview High School
- Participated in curriculum development discussions for advanced biology and chemistry courses

SKILLS & INTERESTS

NHD state qualifier- 1st at school level; 2nd at regionals

AP Scholar with Distinction

National Merit Scholar Semifinalist

National Honor Society member

Concurrent enrollment at FRCC: Economics and Java courses

Laboratory Skills: Scientific methodology, data collection and analysis, scientific writing

Languages: English (Native), Hindi (Native), Spanish (professional fluency)

Research interests: Anatomy & Physiology, Biomedical Engineering, Genetics